

ArchiFlow AI – Reference Architecture Diagram (Numbered)

1. Purpose

This document provides a **single source of truth architecture diagram (logical)** for ArchiFlow AI.

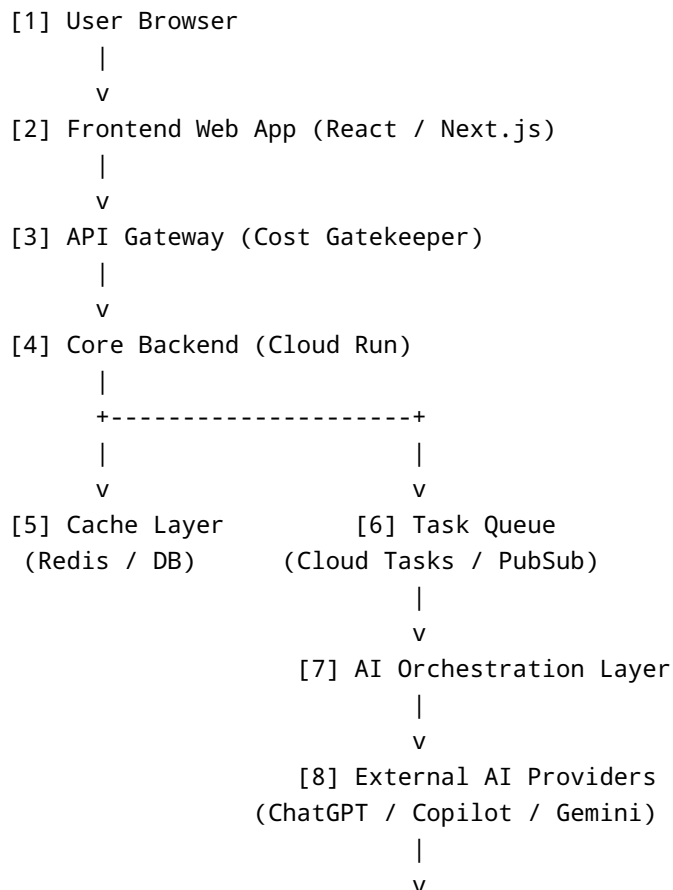
It is designed so that: - Human developers - AI agent coders - Reviewers

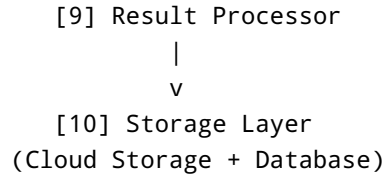
can all reason about the system **without ambiguity**.

This diagram maps **1:1** to the: - Cost-Safe Architecture & Pricing Guardrails - Developer Task Breakdown

2. Architecture Diagram (Logical Flow)

Below is the **logical architecture layout**. Each number represents a mandatory system boundary.





3. Component Legend (MANDATORY BEHAVIOR)

[1] User Browser

- Human user only
- No credentials stored locally
- No AI logic allowed

[2] Frontend Web App

Responsibilities: - UI rendering - Diagram editing interface - File upload (pre-validated)

Hard rules: - Never call AI providers - Never parse documents - Never run long tasks

[3] API Gateway – Cost Gatekeeper

This is the **financial firewall** of the system.

Responsibilities: - Authentication - Subscription validation - Quota validation - Rate limiting - Payload validation

Hard rules: - Reject requests before cost is incurred - No async jobs enqueued if quota fails

[4] Core Backend (Cloud Run)

Responsibilities: - Business logic - Diagram metadata - Job orchestration - Cache lookup

Hard rules: - No direct AI calls - No long-running execution

[5] Cache Layer

Responsibilities: - Store AI responses - Store parsed document outputs

Rules: - Cache key must be deterministic - Cache hit must bypass AI

[6] Task Queue

Responsibilities: - Execute all expensive operations - Control concurrency

Rules: - Max retries: 2 - Max execution time enforced

[7] AI Orchestration Layer

Responsibilities: - Select AI provider - Enforce cost ceilings - Enforce complexity limits

Hard rules: - This is the ONLY place AI calls are allowed

[8] External AI Providers

Providers: - ChatGPT - Microsoft Copilot - Google Gemini

Rules: - Treated as external, metered services - Never directly exposed to frontend

[9] Result Processor

Responsibilities: - Normalize AI output - Sanitize responses - Convert to internal schema

Rules: - No re-triggering AI

[10] Storage Layer

Components: - Cloud Storage (files, exports) - Database (usage, jobs, plans)

Rules: - All outputs persisted - Usage counters updated here

4. Critical Cost-Control Boundaries

Boundary	Protection
[3] API Gateway	Stops quota abuse
[5] Cache Layer	Prevents duplicate AI calls
[6] Task Queue	Controls concurrency

Boundary	Protection
[7] Orchestration	Enforces AI cost ceilings

5. Forbidden Shortcuts (DO NOT IMPLEMENT)

- Frontend → AI Provider
- Backend → AI Provider (direct)
- Sync AI calls
- Unlimited retries
- Background AI loops

Any shortcut above introduces **financial risk**.

6. Instruction for AI Agent Coder

Implement exactly according to the numbered architecture.

If a feature cannot be mapped to a numbered component, it must not be implemented.

Cost safety overrides feature completeness.

7. Final Note

This architecture is optimized for: - USD 20/month pricing - Predictable costs - Safe public launch

Any architectural deviation must be reviewed at product-owner level.